Customer Churn

Data Quality & Exploratory Analysis — Summary Report

Objective: create a reproducible, analysis-ready dataset and document defensible data-quality decisions and initial customer-churn patterns.

Dataset metric	Result
Rows / columns	15 / 11
Missing cells	0
Exact duplicate rows	0
Duplicate CustomerIDs	0
Churned customers	7 (46.7%)
Charge reconciliation	15/15 rows pass

1. Profiling results

The dataset contains 15 customer records and 11 columns. All fields are populated. CustomerID values are unique. Numeric fields have plausible ranges for this sample: Age 23–60, tenure 2–48 months, monthly charges 49.99–149.99, and support tickets 0–6.

2. Cleaning decisions

Check	Result / decision
Text formatting	Trimmed leading/trailing whitespace; category values otherwise preserved.
Data types	Numeric fields explicitly converted; currency fields rounded to 2 decimals.
Missing values	None found; no imputation required.
Duplicates	No duplicate rows or customer IDs; no deletion required.
Outliers	No IQR outliers flagged in numeric columns; retain all records.
Business validation	TotalCharges = MonthlyCharges × TenureMonths for all 15 rows within 0.01.

3. Initial exploratory findings

Overall churn is 46.7% (7 of 15 customers). All Month-to-Month customers in this sample churned, while no One Year or Two Year customers churned. Mean support tickets were 4.29 among churned customers and 1.00 among retained customers. Mean tenure was 7.00 months for churned customers and 29.12 months for retained customers.

Contract-type churn breakdown

Contract type	No churn	Churn
Month-to-Month	0	7
One Year	4	0
Two Year	4	0

4. Validation checks

The cleaned data passed checks for unique non-null customer IDs, zero remaining missing values, non-negative tenure and support-ticket counts, age within 0–120, valid Yes/No churn labels, and charge reconciliation within one cent. The notebook includes executable assertions for these checks.

5. Limitations and responsible interpretation

This is a 15-row synthetic teaching dataset. The observed contract pattern may be an artifact of the sample. These results are descriptive associations, not evidence of causation and not suitable for population-level decisions or model evaluation. A larger, representative dataset is needed.

Prepared as a reproducible task submission. See churn_analysis.ipynb for the full workflow and code.